Assignment :-3

Problem Statement:-

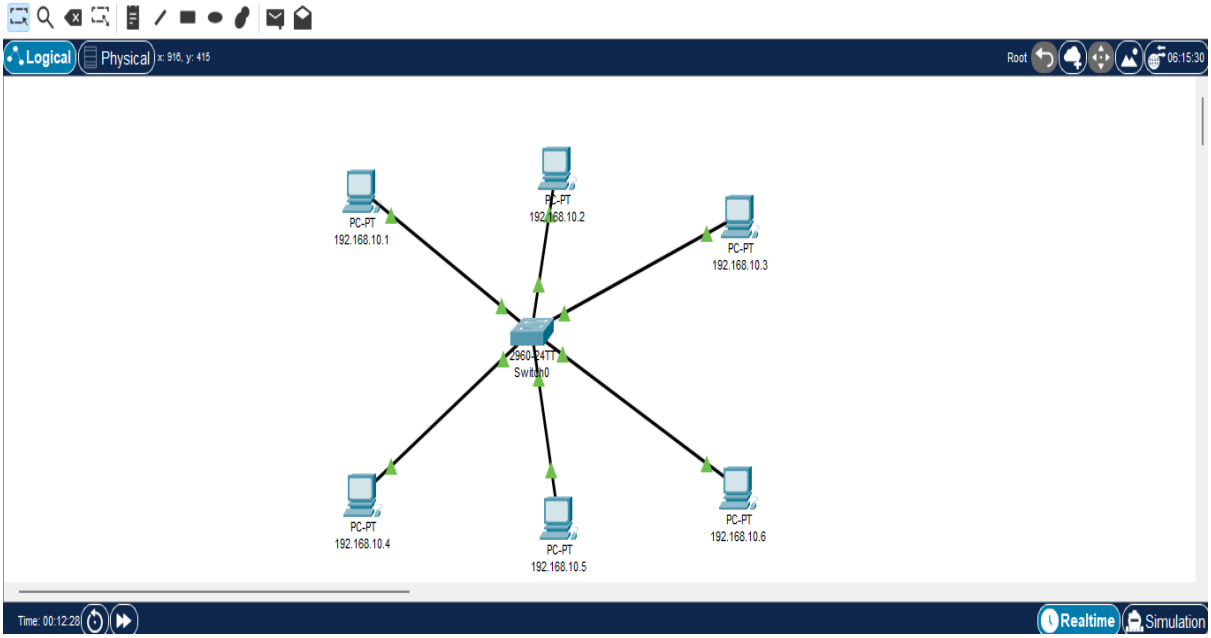
Students will generate simultaneous traffic in a LAN to observe increased delays and packet drops under higher load. The focus is on interpreting simulation behavior rather than complex measurements.

Aim:- Effect of Network Load on Delay and Packet Drops

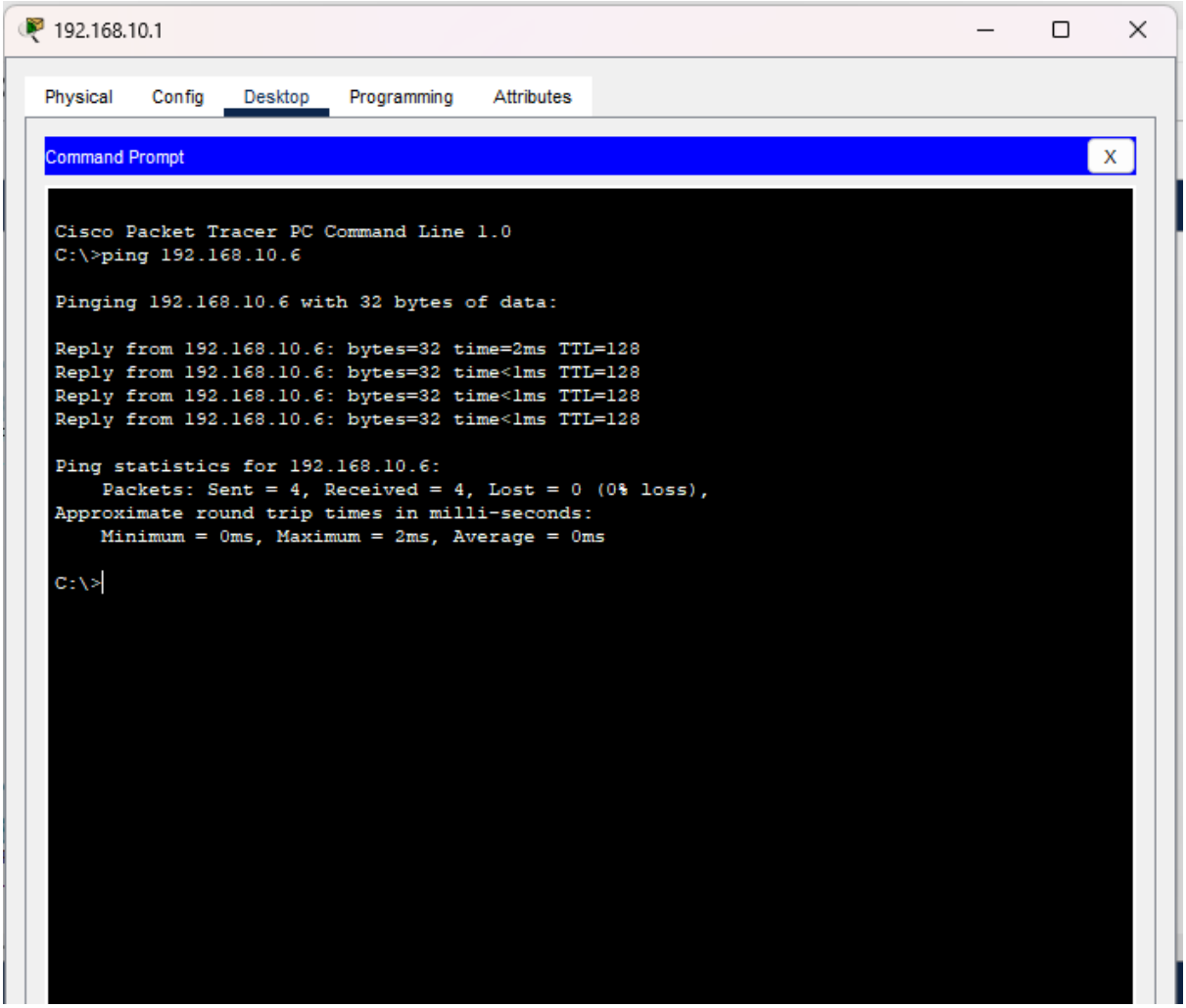
Instructions:-

Task 1: Baseline Test (Low Traffic)

- 1 switch + 6 PCs



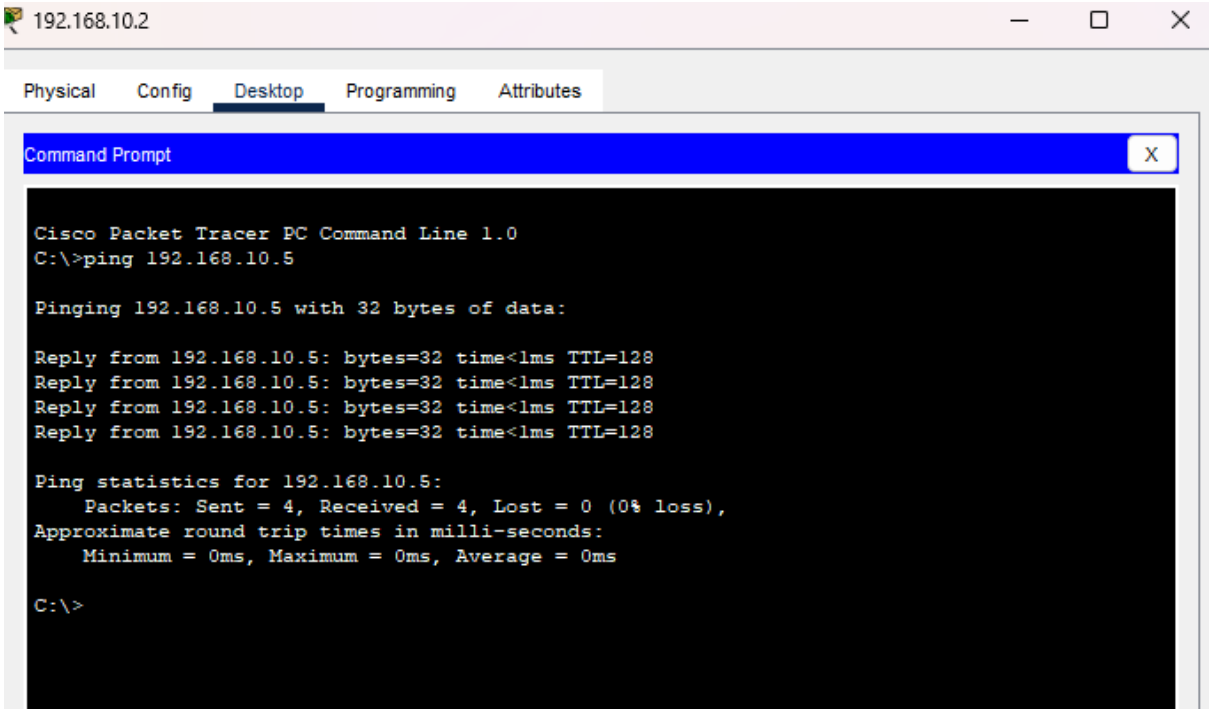
- Ping PC1 → PC6 (single stream)

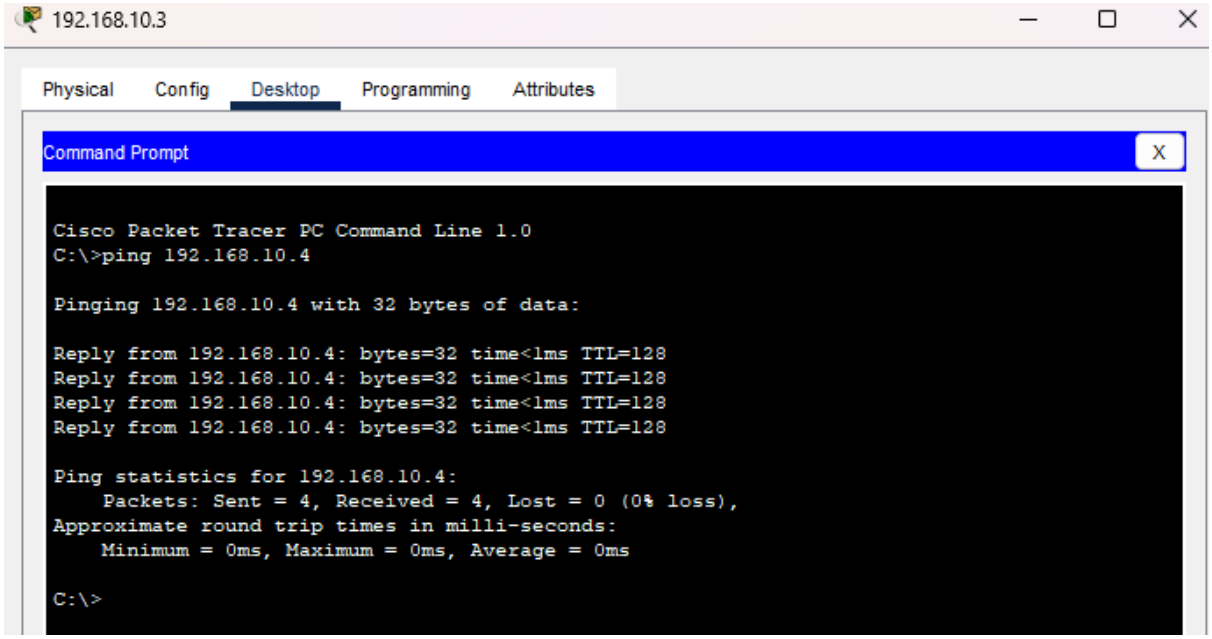


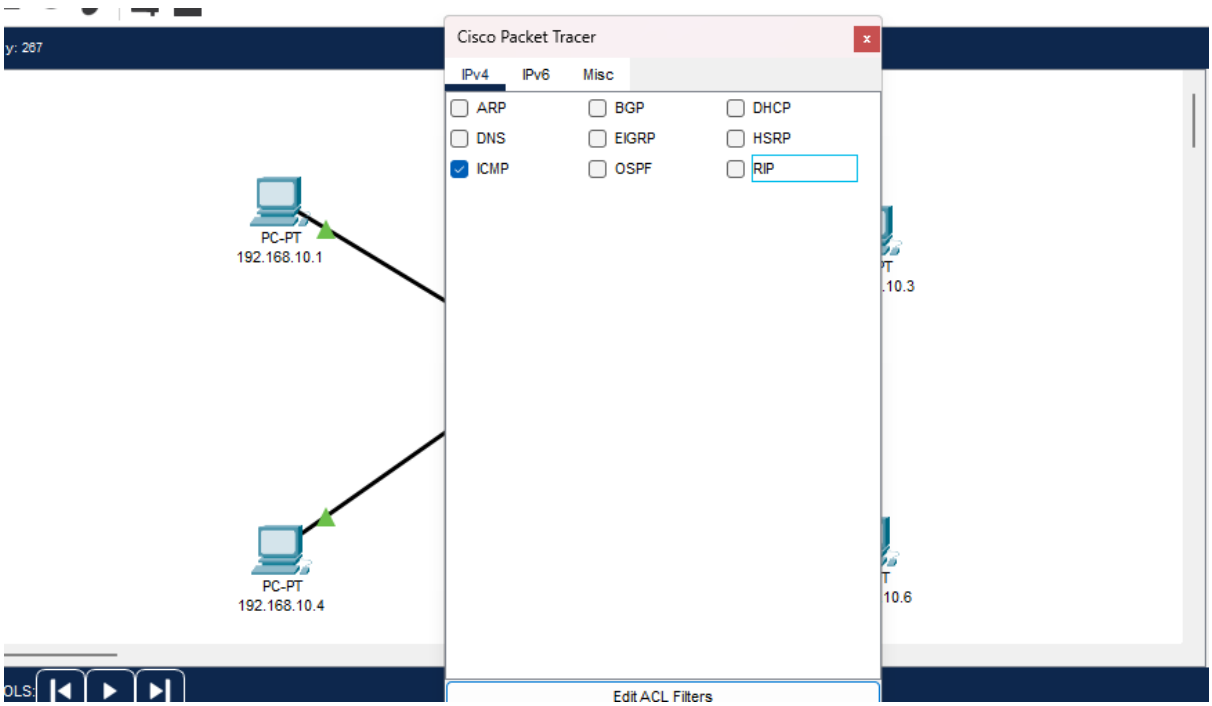
- Record ping success and time behaviour

Task 2: High Traffic Test

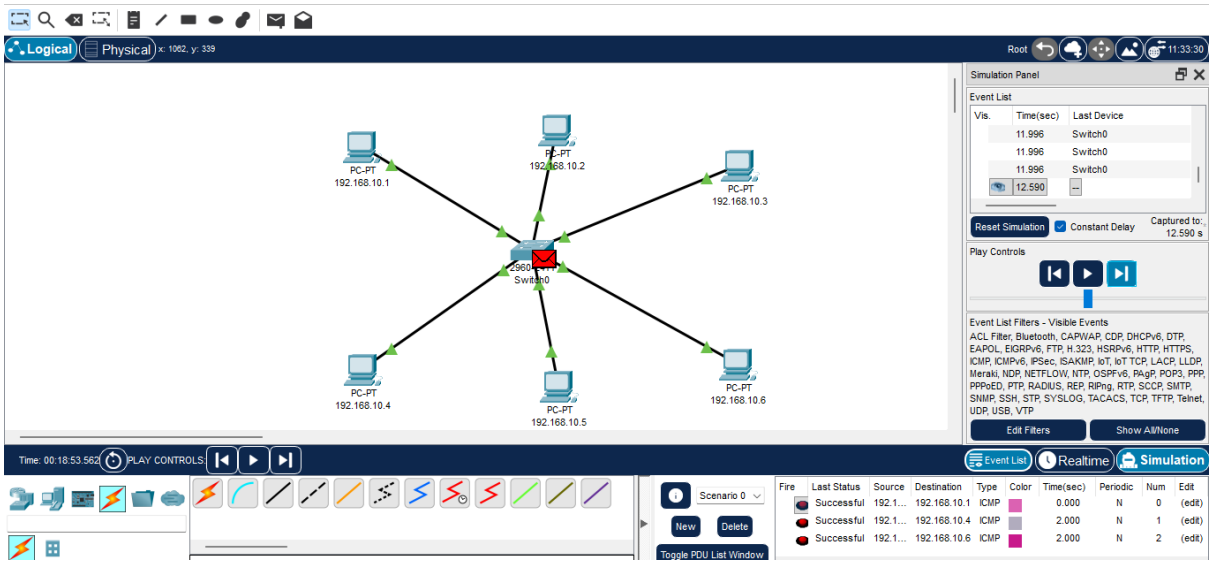
- Start pings from PC1→PC6, PC2→PC5, PC3→PC4 at the same time







- Observe event list and packet timing



Task 3: Observation Summary

1.Multiple ICMP Packets (High Traffic)

- You can see 3 ICMP flows at the same time:
  - PC1 → PC4
  - PC2 → PC5
  - PC3 → PC6

2.Packet Delay (Time Column)

- Look at Time (sec):
  - Packets are not all at `0.000`
  - Some at `2.000 sec`, etc.

3.Switch Behavior (Center Device)

- The switch is handling:
  - Multiple incoming packets
  - Forwarding to correct ports

4.No Packet Loss (Important)

- Bottom panel shows:
  - Successful (ICMP)